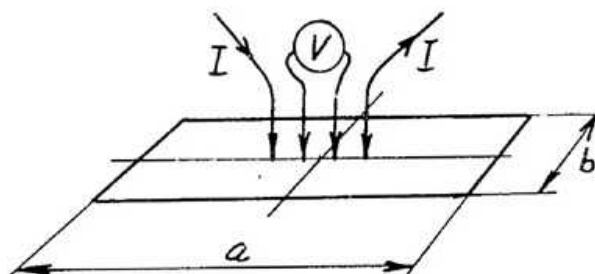


K.2)

Narrow, Rectangular Slice.thickness $t < \frac{s}{2}$

When b is smaller than 3 to 4 times s , it is convenient to express the resistivity in this way:

$$\rho = G \cdot \frac{V}{I}, \quad G = t \cdot \frac{b}{s} \cdot R_2\left(\frac{b}{s}, \frac{a}{b}\right) \quad (26)$$

If $R_2 = 1$, we can write the resistance $\frac{V}{I}$ as

$\frac{V}{I} = \rho \cdot \frac{s}{b \cdot t}$, which is the resistance of a conductor of resistivity ρ , length s and area $b \cdot t$.

So, $R_2 = 1$ corresponds to constant current density in the sample between the voltage probes. As $\frac{b}{s}$ increases, the current density becomes lower far from the probes, and R_2 decreases. R_2 was computed and tabulated by Smits (e). The results are tabulated below and plotted at page 59.

 $R_2\left(\frac{b}{s}, \frac{a}{b}\right)$

$\frac{b}{s}$	$\frac{a}{b} = 1$	$\frac{a}{b} = 2$	$\frac{a}{b} = 3$	$\frac{a}{b} \geq 4$
1			0,9988	0,9994
1,25			0,9973	0,9974
1,5		0,9859	0,9929	0,9929
1,75		0,9826	0,9850	0,9850
2		0,9727	0,9737	0,9737
2,5		0,9413	0,9416	0,9416
3	0,8192	0,9000	0,9002	0,9002
4	0,7784	0,8061	0,8062	0,8062
5	0,7020	0,7150	0,7150	0,7150